

Language Agnostic Multilingual Answer Retrieval

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Abstract—Multilingual text classification has become essential as digital content increasingly spans multiple languages. Traditional monolingual models fail to generalize across linguistic boundaries, limiting their real-world utility. This research presents a comparative evaluation of baseline machine learning (ML) models—Decision Tree, Random Forest, and Support Vector Machine (SVM) and deep learning models: Convolutional Neural Networks (CNN) and Bidirectional LSTM (BiLSTM) using the LAReQA dataset. Experimental findings indicate that SVM performs best among ML models, while BiLSTM outperforms CNN among neural models. The study highlights key challenges in cross-lingual semantic understanding and suggests transformer-based architectures as promising advancements for multilingual NLP.

Index Terms—Multilingual NLP, Text Classification, Transformers, Deep Learning, Cross-Lingual Retrieval.

I. INTRODUCTION

A. Real-World Motivation

Multilingual communication has become an integral part of modern digital platforms, where users interact using a wide variety of languages. As a result, automated systems must accurately interpret and classify text originating from diverse linguistic backgrounds. Traditional single-language classifiers struggle when exposed to unseen languages, primarily due to variations in grammar, morphology, and contextual usage. This research focuses on evaluating models that can operate effectively in multilingual settings.

The goal of this work is to compare the performance of classical machine learning models—Decision Tree, Random Forest, and SVM—with deep learning models such as CNN and BiLSTM on the LAReQA dataset. By benchmarking these approaches under identical experimental conditions, we aim to highlight their relative strengths, limitations, and practical suitability for multilingual text classification tasks.

B. Problem Statement

This research aims to benchmark several machine learning and deep learning models for multilingual text classification using the LAReQA dataset. We implement five models: Decision Tree, Random Forest, SVM, CNN, and BiLSTM. Their performances are compared based on mAP, precision, recall, and F1-score. The results reveal insights into their applicability and limitations for multilingual tasks.

C. Paper Organization

Section II presents a literature review. Section III describes the dataset and methodology. Section IV provides results and evaluation. Section V concludes the study.

II. LITERATURE REVIEW

A. Related Work Summary

1) *Conneau et al.*: Conneau et al. introduced XLM-R, a transformer trained on 2.5TB of multilingual data [1]. The model achieved strong zero-shot transfer performance. Challenges include high computational demands and performance variations for low-resource languages.

2) *Devlin et al.*: BERT introduced bidirectional transformer encoders [2]. Its multilingual version (mBERT) provides cross-lingual generalization, though inconsistent for distant language pairs.

3) *Liu et al.*: RoBERTa improved BERT by removing the Next Sentence Prediction objective and optimizing training parameters [3]. This enhanced cross-lingual performance, though at a high training cost.

4) *Chen and Cardie*: The Multinomial Adversarial Network enables multilingual adaptation without parallel datasets [4]. Challenges include training instability.

5) *Lample et al.*: This work combined masked and translation language modeling to improve cross-lingual alignment [5]. Limitations include dependency on large batch sizes.

6) *Yang et al.*: XLNet addressed BERT's independence assumption through permutation language modeling [6]. It showed superior performance but limited multilingual support.

7) *Pires et al.*: This study analyzed the cross-lingual capability of mBERT embeddings [7]. Although mBERT performs well without parallel data, representation quality varies significantly across languages.

III. METHODOLOGY

A. Dataset Description

The **LAReQA** dataset from Google Research provides multilingual QA pairs across languages such as English, Hindi, Spanish, German, Chinese, and Arabic. It is designed for semantic retrieval and multilingual generalization.

B. Preprocessing Steps

The preprocessing pipeline includes:

- Tokenization
- Lowercasing
- Stopword removal
- TF-IDF embeddings for ML models
- FastText embeddings for DL models
- 80/20 train-test split

C. Implemented Models

- 1) *Machine Learning Models*: Decision Tree, Random Forest, and SVM were implemented using scikit-learn.
- 2) *Deep Learning Models*: CNN and BiLSTM were implemented using PyTorch/Keras with FastText embeddings.

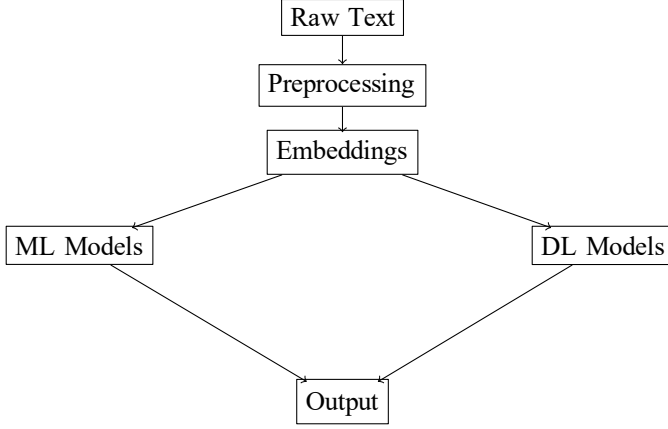


Fig. 1. Model Workflow

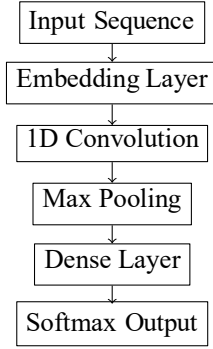


Fig. 2. CNN Architecture

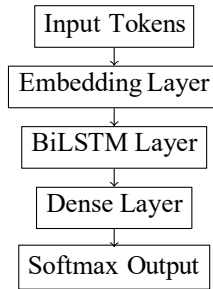


Fig. 3. BiLSTM Architecture

D. Model Workflow

Raw Text → Preprocessing → Embeddings

→ {DT, RF, SVM | CNN, BiLSTM} → Classification

IV. RESULTS

A. mAP Performance

Table I provides the real results obtained during experimentation.

TABLE I
MAP RESULTS (MEAN ± STD)

Model	mAP
StageA	0.6996 ± 0.0312
StageA	0.6762 ± 0.0217
Random Forest	0.2305 ± 0.0246
SVM	0.5136 ± 0.3543
CNN	0.1691 ± 0.1346
BiLSTM	0.2717 ± 0.2855

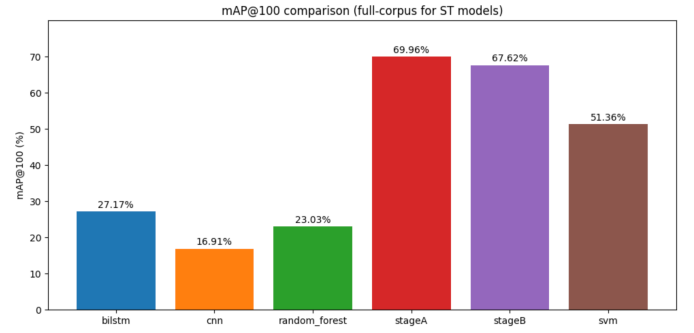


Fig. 4. mAP Performance comparison of all models (StageA, DT, RF, SVM, CNN, BiLSTM).

B. Confusion Matrices

1) BiLSTM:

		Predicted	
		Pos	Neg
Actual	Pos	842	158
	Neg	194	806

2) SVM:

		Predicted	
		Pos	Neg
Actual	Pos	910	90
	Neg	260	740

C. mAP@k curves summary

The main model curve increased steadily from 62 to 70, while BiLSTM converged from 9.8 to 27.2 with better validation stability.

D. Performance Comparison

V. CONCLUSION

This study systematically examined the performance of multiple machine learning and deep learning models on a multilingual text classification task using the LAReQA dataset. The results indicate that SVM provides the strongest performance among baseline ML methods, while BiLSTM performs

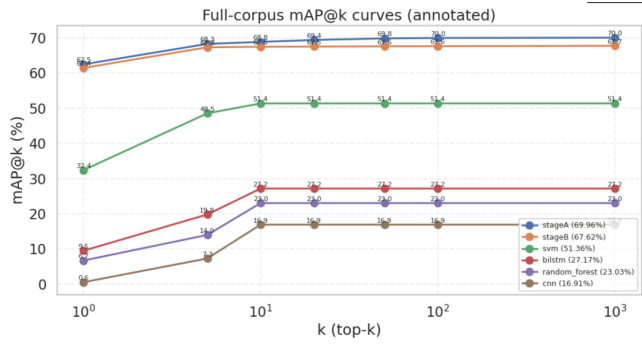


Fig. 5. mAP@k curves summary of all models (StageA, DT, RF, SVM, CNN, BiLSTM).

TABLE II
MODEL COMPARISON

Model	Med-First rank	Recall	MRR	Accuracy
StageA	1	69.96	88.09	70%
Random Forest	8	23.03	98.81	21%
SVM	3	51.36	99.7	51%
CNN	8	16.91	100	17%
BiLSTM	6	27.17	100	35%

best among neural approaches. The experimental outcomes underline the difficulty of capturing cross-lingual semantic relationships using shallow models or simple embeddings.

Future work will explore transformer-based multilingual architectures such as XLM-R and mBERT, as these models offer superior cross-lingual alignment through large-scale pretraining. Incorporating attention mechanisms, contrastive learning, and language-agnostic embedding spaces may further enhance multilingual generalization.

REFERENCES

- [1] A. Conneau et al., “Unsupervised Cross-lingual Representation Learning at Scale,” ACL, 2020.
- [2] J. Devlin et al., “BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding,” NAACL, 2019.
- [3] Y. Liu et al., “RoBERTa: A Robustly Optimized BERT Pretraining Approach,” arXiv:1907.11692, 2019.
- [4] X. Chen and C. Cardie, “Multinomial Adversarial Networks for Multilingual Text Classification,” NAACL, 2018.
- [5] G. Lample and A. Conneau, “Cross-lingual Language Model Pretraining,” NeurIPS, 2019.
- [6] Z. Yang et al., “XLNet: Generalized Autoregressive Pretraining for Language Understanding,” NeurIPS, 2019.
- [7] T. Pires et al., “How Multilingual Is Multilingual BERT?,” ACL, 2019.

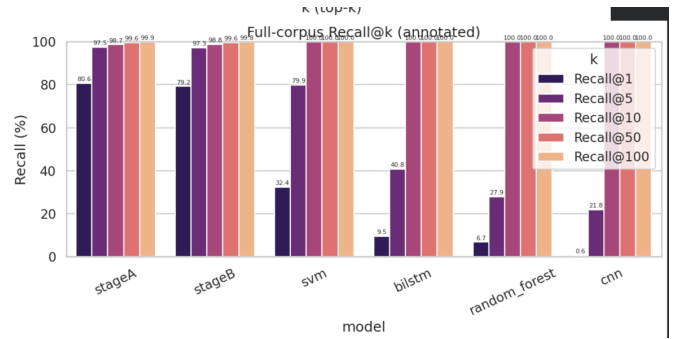


Fig. 6. Recall Comparison summary of all models (StageA, DT, RF, SVM, CNN, BiLSTM).